INCH-POUND

MIL-S-48L
2 August 1990
SUPERSEDING
MIL-S-48K
24 September 1982

MILITARY SPECIFICATION

SOCKS, MEN'S, CUSHION SOLE, STRETCH TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of wool, nylon and cotton, stretch-type, cushion sole socks.

1.2 Classification. The socks shall be in the following classes and sizes as specified (see 6.2):

Class 1 - Black 197
Class 2 - Olive Green 408

Schedule of sizes

Extra extra small	(6 - 6-1/2 - 7)
Extra small	(7-1/2 - 8 - 8-1/2)
Small	(9 - 9-1/2 - 10)
Medium	(10-1/2 - 11 - 11-1/2)
Large	(12 - 13)
Extra large	(14 - 15 and over)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8440

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallet Contractors
MIL-STD-1491	-	Glossary of Knitting Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF AGRICULTURE

Methods of Test for Grades of Wool

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

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3.2 Samples. The socks shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Yarn.

3.3.1.1 Stretch-type knitting yarn. The yarn for knitting the top of foot and leg portion adjacent to the high heel, and for plating the high heel, heel, sole, toe, and ring toe shall consist of a single end of the merino yarn specified in 3.3.1.1.1, twisted or plied with the nylon stretch yarn specified in 3.3.1.1.2, using knitting twist. A stretch core yarn will not be acceptable.

3.3.1.1.1 Merino yarn. The merino yarn shall be 1/30 (worsted count) yarn, made from fleece, pulled sheep's wool, or a combination of both not lower in grade than 56's, U.S. Standard, and cotton, blended in such proportion that the finished yarn contains not less than 50 percent wool on a dry weight basis when tested as specified in 4.4.1. Cotton core yarn will not be acceptable. The merino yarn shall be spun on either the cotton or worsted system.

3.3.1.1.2 Nylon stretch yarn. The yarn shall be 140 denier $\pm$ 5 percent, 2 ply, nylon stretch yarn. Testing shall be as specified in 4.4.1.

3.3.1.2 Top and leg yarn. The yarn for knitting the top and the leg portion above the high heel, shall be a blend of merino yarn spun from a mixture of wool not lower in grade than 56's, U.S. Standard, and cotton. Testing shall be as specified in 4.4.1. This blend shall be in such proportion that the leg portion of the finished sock contains not less than 50 percent wool on a dry weight basis when tested as specified in 4.4.5. Cotton core yarn will not be acceptable. The yarn shall be spun on either the cotton or worsted system.

3.3.1.3 Terry stitch yarn. The yarn for the terry stitch on the inside of the high heel, heel, sole, toe, and ring toe shall be made from wool not lower in grade than 50's, U.S. Standard. The yarn shall be spun on the worsted system. Not finer than 1/16s, 1/18s, and 1/20s or equivalent yarn count shall be used for 108 to 114, 116 to 122, and 124 to 136 needle machines respectively. Testing shall be as specified in 4.4.1.

3.3.1.4 Looping yarn. The yarn for looping the toe of the sock shall be as specified in 3.3.1.1.

3.3.1.5 Seaming thread. The thread for seaming the toes of the socks shall be nylon, natural white, commercial size 4, 1 or 2 ply. The thread shall be 70 denier, and shall have a minimum breaking strength of 0.35 pound and a minimum elongation of 35 percent. Testing shall be as specified in 4.4.1.

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3.3.1.6 Elastic yarn. The elastic yarn (for the rib top) shall be nominal 105 to 110 denier spandex core having a 40 denier raw nylon inner and outer cover. Testing shall be as specified in 4.4.1.

3.4 Color. The color the finished socks shall be as specified (see 1.2 and 6.2). The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished socks shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4.5. When a standard sample is not available, the dyed and finished socks shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.4.5.

3.4.1 Matching. The color of the finished socks shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 $\pm$ 200K, with illumination of 100 $\pm$ 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 $\pm$ 200K. In addition, there shall be no greater variance in shade on the face of the sock between the terried area and the adjacent areas than that exhibited between the terried area and the adjacent area on the inside of the standard sample.

3.4.2. Colorfastness. The dyed socks shall show fastness to perspiration, laundering and crocking equal to or better than the standard sample. In comparing the colorfastness of the standard sample with that of the material under test, specific care will be taken to insure that the same area of both the standard and the test material are taken for testing by any specific test method. When no standard sample is available, the dyed Black 197 socks shall show "good" fastness to perspiration, "fair" fastness to laundering, and shall show an AATCC Chromatic Transference Scale rating for crocking not less than 1.5; and the dyed Olive Green 408 socks shall show "good" fastness to perspiration and laundering, and shall show an AATCC Chromatic Transference Scale rating for crocking not less than 3.5. Testing shall be as specified in 4.4.5.

3.5 Label. Each sock shall have a size and a fiber content marking applied just below the instep and centered between the ankle and toe. The marking shall conform to type III, class 9 of DDD-L-20. The color of the marking medium shall be white. The marking shall be clearly legible and show good fastness to laundering transference (wool method).

3.5.1 Size legend. The size legend shall be printed with characters a minimum of 5/16 inch in height and shall contain the following information as indicated for each size:

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Sock size	Legend
Extra extra small	EXTRA EXTRA SMALL 6 - 6-1/2 - 7
Extra small	EXTRA SMALL 7-1/2 - 8 - 8-1/2
Small	SMALL 9 - 9-1/2 -10
Medium	MEDIUM 10-1/2 - 11 - 11-1/2
Large	LARGE 12 - 13
Extra large	EXTRA LARGE 14 - 15 and over

3.6 Shrink resistant treatment. All of the wool for the finished sock shall be treated for resistance to felting shrinkage in stock, top, yarn or sock form by a controlled oxidation process approved by the contracting activity. The shrink resistant treatment shall not be identified by name or trademark on the socks or on the package.

3.6.1 Alkali solubility. The alkali solubility of the shrink-resistant treated wool for the finished sock, in the form in which it has been treated (stock, top, yarn or sock) shall not be increased by more than 6.0 percent (absolute) over the untreated material when tested as specified in 4.4.1 or 4.4.5.

3.7 Design. The socks shall be seamless, stretch-type, with a true rib-knit top and a plain knit leg and foot with a terry or tuft stitch on the inside of the high heel, heel, sole, toe and ring toe.

3.8 Construction.

3.8.1 Knitting. The socks shall be knitted seamless on a circular machine of not less than 3-1/2 nor more than 4 inches in cylinder diameter with not less than 108 nor more than 136 needles. A minimum of 15 gore needles shall be used in knitting the heel, and a minimum of 15 gore needles shall be used in knitting the toe. The socks shall be knitted so that they will finish to the proper size and length without undue stretching during boarding.

3.8.1.1 Leg. The leg of the sock, down as far as the start of the high heel, shall be plain knit using two ends of the yarn specified in 3.3.1.2.

3.8.1.2 High heel, remaining portion of leg and foot. The high heel, the remaining portion of the leg adjacent to the high heel, and the foot, shall be plain knit with one end of the stretch-type knitting yarn specified in 3.3.1.1. The high heel, heel, sole, toe and ring toe shall be reinforced with a terry stitch thrown to the inside, made with the wool terry yarn specified in 3.3.1.3, in every knitting course of these areas. The terry yarn, for the high heel and sole, shall be laid in at a point not less than 3 needles before the first short butt needle and taken out at a point not less than 3 needles after the last short butt needle in the heel gore. The knitting on all of these needles shall be terried. The terry stitch may be omitted from not more than two courses before the looping course, provided the terry yarn is knit

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with the stretch-type knitting yarn into the knitting of the looper rounds. The two yarns shall be knitted together for at least two courses beyond the looping or loose course.

3.8.1.3 Rib top. The sock shall be a true 1 by 1 ribbed knit using two ends of the knitting yarn specified in 3.3.1.2 and with one end of the elastic yarn specified in 3.3.1.6 laid in every other course, or as an alternate, one end laid in every course. The top of the finished sock shall be finished with a well defined, nonravelling welt, free from loose ends.

3.8.2 Closing the toes. The toes shall be closed by looping as specified in 3.8.2.1 or by seaming as specified in 3.8.2.2.

3.8.2.1 Looped toes. The toes shall be closed with a looper using not less than two ends of the looping yarn specified in 3.3.1.4. The looping line shall be as elastic as the knit fabric and shall be smooth and free from lumps, laps, cuts, and bruises.

3.8.2.2 Seamed toes. The toes shall be closed by seaming using the thread specified in 3.3.1.5. The seam shall be as elastic as the knit fabric and shall be pulled flat to lie smooth and shall be free from lumps, laps, cuts, and bruises. A thread chain of 1/4 to 1 inch shall be left at the ends of the stitching to prevent ravelling.

3.8.2.2.1 Stitch and seam types. The stitch and seam types shall conform to FED-STD-751. The stitch type shall be type 521. The toes shall be seamed inside out with the two edges of the opening superimposed in seam type SSa-1. The sock shall be turned right side out and the seam pulled flat and the edges of the toe opening shall abut as in seam type Fsa-1.

3.8.2.2.2 Stitches per inch. The seams shall be made with 50-60 stitches per inch using the nylon thread specified in 3.3.1.5 in both the needle and looper. Testing shall be as specified in 4.4.5. The stitches-per-inch requirement is based on the socks being in a relaxed state and shall be countered along the left hand row of needle penetrations, that is, the row toward the foot of the sock or away from the edge of the openings as the sock comes off the sewing machine. The socks may be marked off in the relaxed state and the toe stretched to facilitate easier counting.

3.8.3 Length of loose ends. All ends of yarn on the finished sock shall be trimmed. The length of the transfer ends shall be 1 $\pm$ 1/2 inches. The length of the looping ends and free ends at the sole and high heel shall be not less than 1/4 inch nor more than 1 inch.

3.8.4 Finishing. The socks shall be scoured, and boarded on drying forms to their proper shape and size.

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3.8.5 Pairing. The socks shall be matched and paired so that there will be no greater difference in foot length than 1/4 inch and in leg length 1/2 inch. Unless otherwise specified, the use of thread, twine or staples to secure the pair of socks will not be permitted.

3.9 Physical requirements.

3.9.1 Measurements, courses and weights. The measurements, courses and weights of the finished socks shall conform to table I. All measurements in table I shall be made with the sock laid out flat under no tension. Testing shall be as specified in 4.4.5.

TABLE I. Measurements, courses and weights

	See figure 1	Size						
		Extra extra small	Extra small	Small	Medium	Large	Extra large	Tolerance
Leg:								
Number of courses 1/	C	184	184	184	184	184	184	Minimum
Length (inches)	A	15	15	15	15	15	15	Minimum
Foot:								
Number of courses 2/	E	56	72	104	136	180	224	Minimum
Length (inches)	D	5-1/2	7	8-1/2	10	11-1/2	13	-0, +1
Top:								
Number of courses 3/	B	80	80	80	80	80	80	Minimum
Length (inches)	B	4	4	4	4	4	4	±1/2
High heel heights (in)	F	4	4	4	4	4	4	Minimum
Ring toe width (in)	G	1	1	1	1	1	1	Minimum
Conditioned weight per dozen pairs (oz)		26	27.5	30	32.5	35.5	38.0	Minimum

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1/ Leg courses shall be counted from first short butt heel gore needle to bottom edge of ribbed top.

2/ Foot courses shall be counted from first short butt toe gore needle to first short butt heel gore needle.

3/ Top courses shall be counted from bottom edge of welt to transfer or joining point of leg fabric.

3.9.2 Stretch. The finished sock shall conform to the requirements listed in table II when tested as specified in 4.4.5.

TABLE II. Stretch requirements

	Size						
	Extra extra small	Extra small	Small	Medium	Large	Extra large	Tolerance
Crosswise stretch (in):							
Leg 1/	7	7	7	7	7	7	Minimum
Foot 2/	7	7	7	7	7	7	Minimum
Lengthwise stretch (inches):							
Foot	8	9.5	11	13	15.5	18.5	Minimum

1/ Measure across a point between the bottom of heel and bottom of rib top.

2/ Measure across a point midway between heel and toe.

3.9.3 Bursting strength. The leg fabric above top of extended heel of the finished sock shall have a bursting strength of not less than 70 pounds and the reinforced (terry stitch) sole fabric of the finished sock shall have a bursting strength of not less than 125 pounds when tested as specified in 4.4.5.

3.9.4 Nonfibrous material content. The starch and protein content, including chloroform-soluble and water-soluble material, of the finished sock, shall not exceed 3.0 percent when tested as specified in 4.4.5.

3.9.5 Thickness. A double thickness of the terry portion (terry to terry) of the finished sock shall have a minimum thickness of 0.115 inch when tested as specified in 4.4.5.

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3.9.6 Dimensional stability. The finished sock shall be tested for felting shrinkage or elongation in foot length and leg length as specified in 4.4.5. At each of the specified locations, the felting shrinkage or elongation shall be not more than 6.0 percent.

3.10 Workmanship. The finished socks shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

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4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, testing shall be performed on the components listed in table V. The methods of testing specified in FED-STD-191 wherever applicable, and as specified herein shall be followed. The sample unit and the basis for expressing lot size shall be as specified in table III. Except where otherwise specified (see 4.4.5), the sample size shall be in accordance with table IV. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final result.

TABLE III. Sample unit and basis of lot size

Component	Basis of lot size	Sample unit
Wool		
Stock	Pound	1/4 pound 1/
Top	Pound	1 yard 1/
Yarn	Spool, cone or tube	1 spool, cone or tube 1/
Yarn	Spool, cone or tube	1 spool, cone or tube

1/ An equal quantity of untreated material is required for the alkali solubility test.

TABLE IV. Sampling for testing

Lot size	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE V. Component and material tests

Material	Characteristic	Requirement paragraph	Test method
Merino yarn	Count	3.3.1.1.1	4021
	Wool grade	3.3.1.1.1	4.4.2
	Wool content	3.3.1.1.1	2100
	Cotton - material identification	3.3.1.1.1	1200 1/
Nylon stretch yarn	Material identification	3.3.1.1.2	1530 1/
	Denier	3.3.1.1.2	4021 1/
	Ply	3.3.1.1.2	Visual 2/
Top and leg yarn	Wool grade	3.3.1.2	4.4.2
	Cotton - material identification	3.3.1.2	1200 1/
Terry stitch yarn	Wool grade	3.3.1.3	4.4.2
	Wool content	3.3.1.3	2101
	Count	3.3.1.3	4021
Nylon seaming thread	Material identification	3.3.1.5	1530 1/
	Ply	3.3.1.5	Visual 2/
	Denier	3.3.1.5	1/
	Breaking strength	3.3.1.5	4100
	Elongation	3.3.1.5	4100
Elastic yarn core	Material identification	3.3.1.6	1/
	Denier	3.3.1.6	1/
Cover	Material identification	3.3.1.6	1/
	Denier	3.3.1.6	1/
Wool	Alkali solubility	3.6.1	2800 3/

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ One determination per sample unit and the results reported as pass or fail.

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3/ The wool shall be tested for alkali solubility in the form in which it was treated and the sample unit shall consist of both treated and untreated material.

4.4.2 Determination of wool grade.

4.4.2.1 Stock form. The wool shall be visually examined for grade (see 3.3.1) in grease or scoured and untreated form (prior to picking or carding) by comparison with the applicable U.S. Standard. In the event of dispute resulting from the above comparison, the wool grade shall be determined by the width method (wedge) approved by the Government. In the event of dispute, the examination for grade shall be performed on a composite sample of 1 pound of wool from each 10,000 pounds or fraction thereof in the lot. The composite sample shall be selected from not less than 10 percent of the bales or boxes in the lot. The lot shall be unacceptable if the sample does not conform to the wool grade requirements specified.

4.4.2.1.2 Top form. When examined in top form, the grade shall be determined by comparison with the applicable U.S. Standard. In the event of dispute resulting from the above examination, the wool grade shall be determined by the width method (wedge) approved by the Government. The sample unit shall be 1 yard of untreated top. The results of each sample unit shall be reported separately.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table VI. The lot size shall be expressed in units of socks. The sample unit shall be one sock and the selection shall be by pairs. The inspection level shall be I and the acceptable quality level (AQL), expressed in defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects. Applicable material knitting defects shall be as defined in MIL-STD-1491.

TABLE VI. End item visual defects

Defect	Classification Major	Minor
Any hole, run, or dropped stitch, including hole in looping	101	
Thin area, mend, or split stitch	102	
Welt omitted, cut, frayed, or ravelled	103	
Welt uneven, or not clearly defined		201
Poor looping resulting in looping lines less elastic than knit fabric, lumps, and dog ears--will not level out. Less than two ends of looping yarn	104	

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TABLE VI. End item visual defects (cont'd)

Defect	Classification Major	Minor
Loose end:		
Trimmed less than specified	105	
Trimmed longer than specified		202
Ends floating more than 1 inch on inside of sock		203
Any floated end on outside of sock	106	
NOTE: Floated ends which at one point are firmly attached as a part of the knitted fabric, with the other end loosely caught in the stitch shall be regarded as loose ends		
Pattern of knit not as specified throughout entire sock	107	
Ring toe omitted	108	
Terry stitch omitted:		
For three or more complete courses	109	
For an area exceeding 2 inches in combined length and width	110	
Nonuniform in overall appearance, not well shaped		204
Slubby yarn more than three times normal diameter		205
Streak, stain, shade bar, or spot, clearly visible 1/		206
Rancid or bad odor		207
Off-shade		208
Two tone color contrast between outside of terried areas and adjacent areas of sock exceeds approved limits		209
Mismatched for shade		210
Vegetable matter clearly noticeable		211
Any burn or scorch in fabric	111	
Marking missing, incorrect, incomplete, illegible, or numerals of improper dimensions		212
Preshrinking process identified by name or trademark		213

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TABLE VI. End item visual defects (cont'd)

Defect	Classification Major	Minor
Applicable to seamed toes:		
Wrong seam or stitch type	112	
Thread chain less than 1/4 inch but not less than 1/8 inch		214
Thread chain more than 1 inch		215
Thread chain less than 1/8 inch	113	
Ends of stitching more than 1 inch, not chained off		216
Ends of stitching less than 1/2 inch, but not less than 1/4 inch, not chained off		217
Ends of stitching less than 1/4 inch, not chained off	114	
Skipped stitches, run offs, or broken thread resulting in any open seam	115	
Edges of toe opening do not abut		218
Seam not flat or smooth (i.e., lumps or laps, tightness or waviness)		219
Tuck stitches on outside of sock in terried area:		
Two or more floating over more than two courses		220
Any clearly visible accumulation of tuck stitches covering area of more than 2 inches in its longest dimension		221

1/ At normal inspection distance (approximately 3 feet).

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table I. Any dimension not within the specified tolerance shall be classified as a defect. In addition, any pair of socks mismated for foot length by more than 1/4 inch or leg length by more than 1/2 inch shall be classified as a single defect. The lot size shall be expressed in units of socks. The sample unit shall be one sock and the selection shall be by pairs. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end items shall be tested for the characteristics listed in table VII. The methods of testing specified in FED-STD-191 wherever applicable, and as listed in table VII shall be followed. Except for colorfastness, weight, stretch tests and dimensional stability, the sample size shall be in accordance with table IV. The sample size for colorfastness, weight, stretch tests and dimensional stability shall be one sample unit. All test reports shall contain the individual values utilized in expresssing the final result. The lot size shall be expressed in units of one sock each. The sample unit and the disposition of the lot shall be as follows:

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Characteristics	Sample unit	Disposition of lot: the lot shall be unacceptable if:
Weight	One dozen pairs of socks from each size represented in lot.	The sample unit for any size fails to meet the requirement specified.
Dimensional stability and stretch	Sixteen individual socks from each size represented in lot.	The sample unit for any size fails to meet any requirement specified.
Colorfastness	Two individual socks from each size represented in lot.	One or more socks fail to meet any requirement specified.
All remaining test characteristics	Twelve individual socks drawn in the same proportion as represented in the lot, plus one untreated sock which shall be the same size as one of the treated socks (to be used for alkali solubility comparison when treated in sock form).	One or more socks fail to meet any requirement specified.

Table VII. End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as
Wool content of top and leg portion	3.3.1.2	2100		
Labile sulfur	3.4	2020		
Colorfastness to:				
Laundering	3.4.2	5614		
Crocking	3.4.2	5651		
Perspiration	3.4.2	5680		
Alkali solubility (when treated in sock form)	3.6.1	2800		

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Table VII. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as
Seamed toes:				
Stitches per inch	3.8.2.2.2	Count	1	Actual count
Number of courses:				
Leg	3.9.1	Count	1	Actual count
Foot	3.9.1	Count	1	Actual count
Top	3.9.1	Count	1	Actual count
Weight per dozen pairs (conditioned)	3.9.1	Scale	1	To nearest 0.1 ounce
Stretch (crosswise):				
Leg	3.9.2	1/	8	Average of 8 determinations to nearest 0.1 inch
Foot	3.9.2	1/	8	Average of 8 determinations to nearest 0.1 inch
Stretch (lengthwise):				
Foot	3.9.2	2/	8	Average of 8 determinations to nearest 0.1 inch
Bursting strength:				
Leg	3.9.3	5120		
Sole	3.9.3	5120		
Nonfibrous material	3.9.4	2611 3/		
Thickness	3.9.5	5030 4/	10	Average of 10 determinations to the nearest 0.001 inch
Dimensional stability	3.9.6	7651 5/		

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1/ The stretch measurement shall be evaluated on a Houchin modification of a Schieffer machine described in Method 7540 of FED-STD-191. The sock shall be pulled over the form so that the 2 inch edges shall be parallel to the foot of the sock. The indicator pin shall be inserted in the upper 2 inch edge of the separating form and a 10 pound weight slowly applied so as to cause separation of upper and lower slotted form with resultant stretch of the sock under a uniform load over the 2 inch edges. The socks shall be evaluated after the weight has hung freely for 30 seconds and shall be expressed in 0.1 inches as the distance between the upper and lower edges of the separated form.

2/ The lengthwise stretch of the socks shall be measured using the sock measuring machine apparatus and measurement of specimen procedure specified in Method 7560 of FED-STD-191. The measurement shall be taken to the nearest 0.1 of an inch and the measured stretch "foot length" obtained in accordance with this procedure shall be considered as the lengthwise foot stretch.

3/ Specimen shall be cut from the leg portion.

4/ The presser foot shall be circular with a diameter of 0.375 inch $\pm$ 0.001 inch and with the moving parts connected therewith weighted so as to apply a total load of nine pounds per square inch. Two readings shall be taken on each of five specimens. The thickness of the sample unit shall be the average of ten results obtained from the specimens tested and shall be reported to the nearest 0.001 inch.

5/ In the felting shrinkage procedure, the period of laundering in the wash wheel shall be 2 hours. Elongation shall be prefixed with a minus (-) sign.

4.4.6 Packaging examination. The fully packaged socks shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application Preshrinking process is identified by name or trademark
Materials	Any component missing, damaged, or not as specified

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Examine	Defect
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of container flap, taping, loose strapping, improper taping, or inadequate stapling Bulged or distorted container
Content	Number of bundles per container is more or less than specified Number of pairs of socks per bundle is more or less than required 1/ Size shown on one or more socks not as specified on shipping container 1/

1/ For this defect, two bundles shall be examined from each shipping container in the sample.

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each pair of socks shall be folded at the heels. Four pairs of class 1 socks of one size only, or five pairs of class 2 socks of one size only, shall be evenly stacked in a bundle with every other pair reversed. Each bundle shall be tied around each end with cotton tape or twine to form a unit pack.

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5.1.2 Commercial preservation. Socks shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Class 1 or class 2 socks of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Quantities and packing arrangements for level A unit packs shall be as shown in table VIII. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15-1/2 inches in width and 6-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Four ounces of napthalene flakes shall be uniformly distributed throughout the contents of each container (see 6.5). Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2 Level B packing. Class 1 or class 2 socks of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Quantities and packing arrangements for level A unit packs shall be as shown in table VIII. Inside dimensions of each container shall approximate 23-1/2 inches in length, 15-1/2 inches in width and 6-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Four ounces of napthalene flakes shall be uniformly distributed throughout the contents of each container (see 6.5). Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

TABLE VIII. Quantities and packing arrangements

Class and size	Quantity per shipping container (pairs)	Packing arrangement for level A unit packs
Class 1:		
Extra-extra-small through Large	72	9L x 1W x 2D
Extra-large	48	6L x 1W x 2D

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TABLE VIII. Quantities and packing arrangements (cont'd)

Class and size	Quantity per shipping container (pairs)	Packing arrangement for level A unit packs
Class 2:		
Extra-extra-small through Large	70	7L x 1W x 2D
Extra-large	50	5L x 1W x 2D

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Socks, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), socks, packed as specified in 5.2.2 and 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The socks are intended for use by male personnel of the Department of Defense.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Class and size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting agency issuing the invitation for bids or request for proposal.

6.5 Napthalene flakes. The napthalene flakes requirement is not applicable when Government furnished material of wool or part wool content has been mothproof treated (see 5.2.1 and 5.2.2).

6.6 Dye formulations. The socks have been successfully dyed using the following formulations (see 3.4):

Black - 197
Direct Black - WNC
Acid Yellow - 63-13095
Acid Yellow - 65-14170

OG - 408
Direct Orange - 72
Direct Black - 6
Direct Brown - 58-22340
Direct Black - 51-27720
Acid Red - 3NB
Acid Yellow - 65-14170
Acid Blue - NGB

6.7 Subject term (key word) listing.

Cotton
Footwear
Nylon
Wool

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD, EA
Air Force - 11, 82
DLA - CT

User activities:

Navy - MC
Air Force - 45

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-48L	2. DOCUMENT DATE (YYMMDD) 1990 August 2

3. DOCUMENT TITLE
SOCKS, MEN'S, CUSHION SOLE, STRETCH TYPE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial: 508-651-5221 (2) AUTOVON: 256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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